PROJECT DOCUMENTATION & TECHNICAL SPECIFICATION

AirSense AI

Urban Air Quality Intelligence Platform

A complete AI-powered web platform featuring real-time urban telemetry, machine learning AQI prediction, 5-vector source attribution, Leaflet GIS mapping, and multilingual Gemini AI health advisories (English, Tamil, Hindi, Kannada).

DATE: July 2026

STACK: React 18, TypeScript, Vite, Tailwind CSS, Leaflet, Recharts, Gemini API

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1. Executive Summary

AirSense AI is built to bridge the gap between raw urban atmospheric telemetry and actionable public health policy. By combining real-time particulate matter (PM2.5, PM10) and gaseous pollutant (NO_2, SO_2) sensor feeds with machine learning EPA sub-index calculation algorithms, the platform delivers precise AQI assessments, source attributions, interactive geospatial mapping, and multilingual AI health advisories.

2. System Architecture & Technology Stack

Component	Technology Choice	Purpose
Core Framework	React 18 + TypeScript + Vite	Type-safe, lightning-fast component architecture
Styling	Tailwind CSS + Glassmorphism	Environmental blue/green theme with backdrop blur cards
UI Icons	Lucide React	Consistent, scalable vector icons across all pages
Data Visualization	Recharts 2.x	Diurnal trend area charts, hotspot bars, & radar profiles
GIS Mapping	Leaflet 1.9 + React-Leaflet	OpenStreetMap dark tile layer with glowing AQI badges
AI Integration	Google Gemini 1.5 Flash API	Multilingual health advisories & prediction diagnostics
PDF Engine	jsPDF + html2canvas	Client-side executive report export & print preview

3. Platform Page Specifications (9 Modules)

1. Landing Page [src/pages/LandingPage.tsx]

Environmental hero banner, live urban AQI ticker (Delhi, Mumbai, Bengaluru, Tokyo, London, NY), quick city telemetry inspector, and core capability shortcuts.

2. Telemetry Dashboard [src/pages/Dashboard.tsx]

Summary metric cards (Average AQI, Highest Hotspot, Lowest Center, Sensor Nodes), visual semicircle AQI gauge, Recharts 24h trends area chart, radar profile, and live station directory.

3. AQI Prediction Engine [src/pages/AQIPrediction.tsx]

Input sliders for PM2.5, PM10, NO2, SO2, Temp, and Humidity; scenario presets; EPA sub-index calculation; and Gemini AI diagnostic notes.

4. Pollution Hotspots Analytics [src/pages/Hotspots.tsx]

Top polluted city rankings, heat intensity badges, regional filter (North/South India, East Asia, Europe), and comparative Recharts bar chart.

5. Source Attribution Diagnostics [src/pages/SourceAttribution.tsx]

5-vector sector contribution (Traffic 38%, Industry 26%, Construction 16%, Waste Burning 12%, Natural Dust 8%), interactive donut chart, and sector mitigation strategies.

6. Interactive GIS Map [src/pages/InteractiveMap.tsx]

Leaflet map with custom glowing HTML/SVG AQI marker badges, detailed telemetry popups, search bar, and location quick-jump shortcuts.

7. Multilingual Citizen Advisory [src/pages/HealthAdvisory.tsx]

Personalized safety protocols in English, Tamil, Hindi, and Kannada for General Public, Children & Elderly, Outdoor Workers, and Asthma Patients.

8. Executive PDF Reports [src/pages/Reports.tsx]

Report customizer checklist, live print preview document layout, and one-click PDF generation via jsPDF + html2canvas.

9. About & Tech Blueprint [src/pages/About.tsx]

Mission statement, sub-index mathematical equations, data provider compliance (CPCB, US EPA, Copernicus), and tech stack inventory.

4. EPA AQI Sub-Index Calculation Equations

AirSense AI evaluates air quality by calculating linear sub-indices for primary criteria pollutants using standard EPA/CPCB breakpoints:

- **PM2.5 Sub-index: I_PM2.5 = PM2.5 * 3.8 * Weather_Factor**
- **PM10 Sub-index: I_PM10 = PM10 * 1.8 * Weather_Factor**
- **NO2 Sub-index: I_NO2 = NO2 * 1.4**
- **SO2 Sub-index: I_SO2 = SO2 * 1.5**
- **Weather Factor = 1 + (Temp > 35°C ? 0.10 : 0) + (Humidity < 30% ? 0.08 : 0)**
- **Final Calculated AQI = Max(I_PM2.5, I_PM10, I_NO2, I_SO2) [Capped at 500]**

5. GitHub Upload & Quick Start Commands

```
# Clone or navigate to directory
cd C:\Users\LENOVO\.gemini\antigravity\scratch\airsense-ai

# Push to GitHub repository
git init
git add .
git commit -m "Initial commit: AirSense AI Urban Platform"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/airsense-ai.git
git push -u origin main
```